Dishadharti — ML Technical Report

Everything the ML track built: inputs, data, methods per factor, measurements, decisions, sources with links, and how to run and reuse it. Written so every teammate — technical or not — knows exactly what is done. Every number below was read off a real run.

---

1. Problem and use cases

A municipal officer receives a development proposal and must decide: approve, condition, or reject. Today that decision has no climate evidence. This system scores the proposal on flood, heat, green cover, and embodied carbon (0–100, high = worse), explains the scores in plain language, and attaches redesign conditions whose effects are measured by re-scoring — not asserted.

Use case	Who	What the system does
Pre-approval screening	Municipal officer	Pin plot + two areas + bill of quantities; receive four scores, a verdict (approve / condition / reject), and ranked conditions.
Redesign guidance	Builder / architect	Each condition shows its measured score delta, so redesign targets the factor that moves most.
Tree compliance	Tree Officer	Plot-level tree ledger: what stays, what falls, age-equivalent compensation as a range.
Spoken explanation	Officer / public	Chatbot narrates the computed JSON. It reads numbers; it never computes or adjusts them.
Reuse in another city	Any municipal team	The engine holds no city knowledge: new config + cube + census + wards, same scorer.

2. Inputs: what the builder hands over

Four things, nothing else trusted. Storeys and project type are framing words only — they never change a number.

Input	Example	How it is used
Site boundary	Map pin / GeoJSON / KML polygon	Decides the ward and exactly which census trees stand inside the footprint (point-in-polygon, never a ward-id join).
Built-up area	3,000 m2	Drives flood (impervious fraction) and divides carbon intensity. Must not exceed plot area — rejected otherwise.
Plot area	8,000 m2	The site extent. Canopy percentage divides by this; zero or missing plot fails validation with reasons.
Bill of quantities	cement 120,000 kg; steel 45,000 kg	Each line times its emission factor. 102 name aliases resolve real-world spellings; over 10% unrecognised mass withholds the carbon score instead of guessing.

3. System architecture

Two phases. Phase 1 runs once, offline: satellites and census files are baked into two artefacts — epoch_cube.json (864 one-km Pune cells: land cover, day/night surface temperature, elevation) and tree_index.sqlite (4,009,623 deduplicated census trees, spatially indexed). Phase 2 runs per proposal in ~10 seconds with no internet: point-in-polygon on 864 rows plus a tree-ledger walk, then deterministic arithmetic. This is a lookup table, not retrieval: no embeddings, no vector store. A committee can audit an exact number and challenge it.

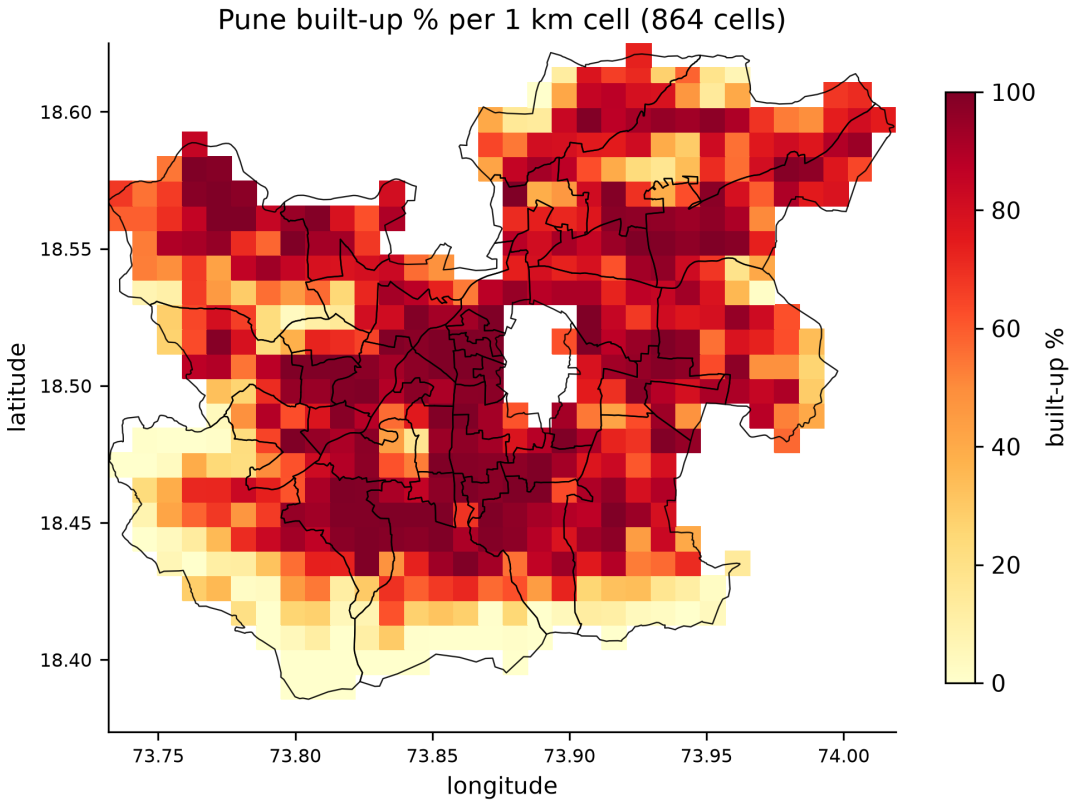


Figure 1. Built-up share per cell with ward boundaries. Every flood score in the system comes out of this partition.

4. Data built

Artefact	Content	Count / status
Epoch cube	2024 land cover + temperatures + elevation	864/864 cells complete
Tree index	Deduplicated 2019 census, spatial lookup	4,009,623 trees
Rainfall series	124 annual 1-day maxima, IMD grids	imd_pune_annual_max.csv
Emission table	38 materials, 102 aliases, India factors	materials.json
Species table	Census species data, full-index audit	species.json

The partition was the hard part. An early build silently deleted every water pixel, read vegetation from the wrong sensor at half its value, and smeared classifier uncertainty into phantom snow. All three were found by measurement and gated: the cube now refuses to build unless its nine land classes sum to one per cell. Water went 0 to 258 cells; vegetation median 6.3% to 30.2%.

5. Flood

Standard SCS curve-number hydrology (NRCS TR-55): area-weighted coefficients versus a vegetated baseline, excess runoff at the design storm. Drainage proximity can only lower the score; natural streams never count. Design storm 214 mm: Gumbel 100-year quantile over 124 IMD annual maxima (fit p = 0.95, interval 191–234). Across 41 wards: 33.9–53.3; built-up predicts it at r = +0.92, vegetation at r = –0.95. Slope band is currently unavailable (cube has no mean_slope_deg).

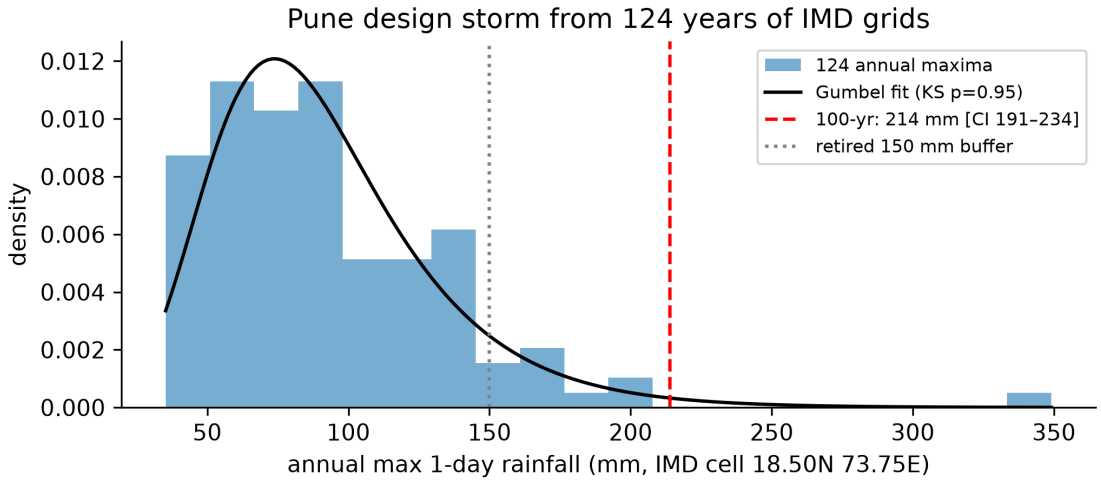


Figure 2. Gumbel fit to 124 IMD annual maxima. The retired 150 mm buffer sat between the 10- and 25-year storms.

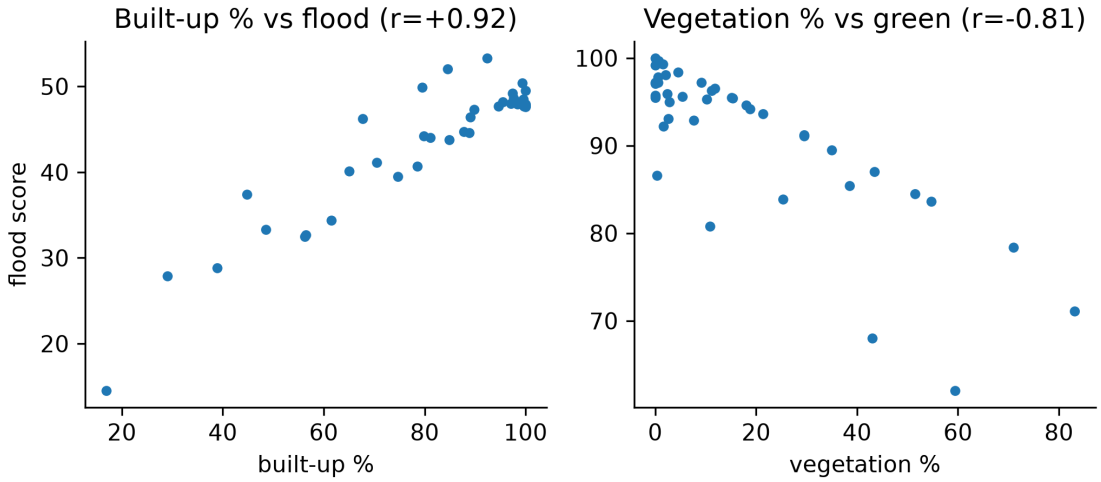


Figure 3. Flood rises with concrete, green cover falls with vegetation loss. Both point the right way.

6. Heat

Night surface temperature minus a rural vegetated ring, over a measured 5.0 C span. City mean +1.62 C above rural (Yale YCEO anchor sampled live at +1.139). Daytime was tested twice and retired: 11am contrast +0.11 C with seasonal sign flip; 1:30pm peak-hour contrast –0.14 C (dry rural soil out-heats shaded city — documented dryland effect). Reports label heat night-only and carry unscored afternoon context instead of a faked score. Spread 32.8–98.1, the widest of any factor.

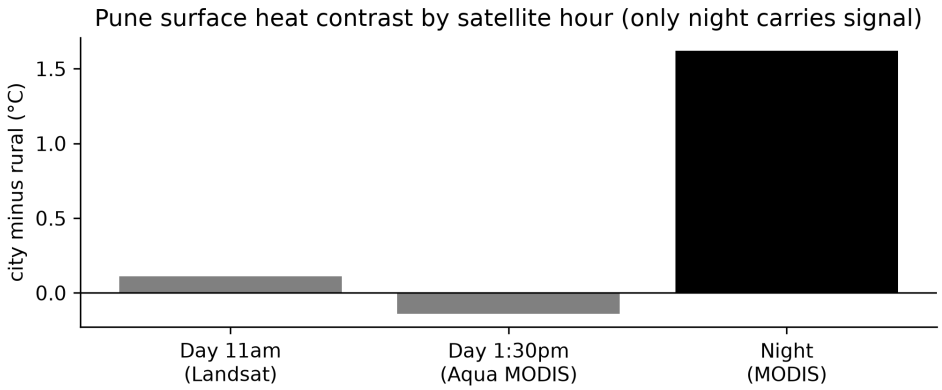


Figure 4. Only the night column carries signal.

7. Green cover and carbon

Green: census canopy (70%) + satellite ground vegetation (30%), stratified against double-counting; missing data returns unavailable, never perfect. Compensation is age-equivalent ranges under the 1975 Trees Act as amended 2021. Spread 73.2–93.0, r = –0.81. Carbon: bill of quantities times India factors (cement 0.91, steel 2.6, brick 0.39; aluminium 20.88), per square metre against a 1200 scale. Above 10% unknown mass the score is withheld. Flat across wards by design (identical BOQ); varies across sites (26.3–84.5).

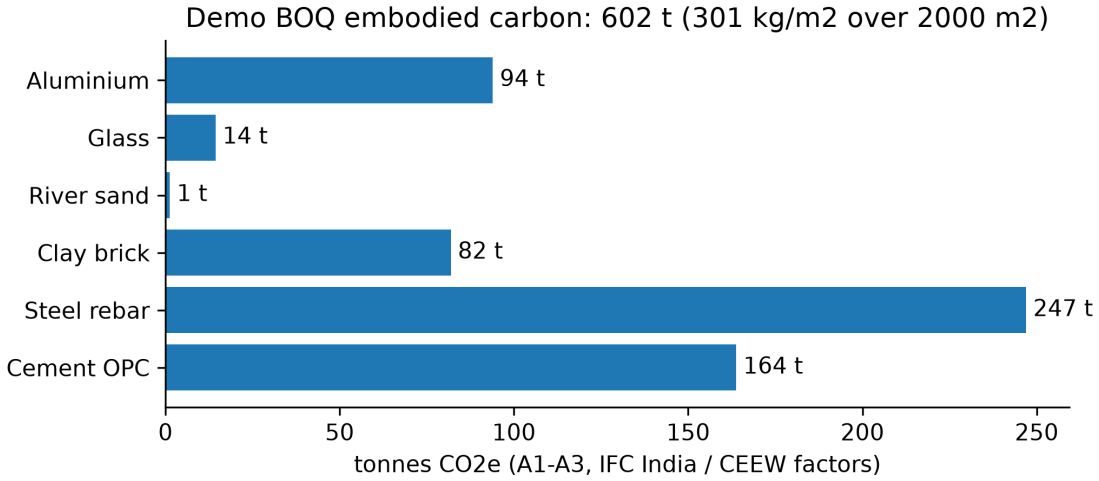


Figure 5. Demo bill: 602 tonnes, steel first. Every bar multiplies out by hand.

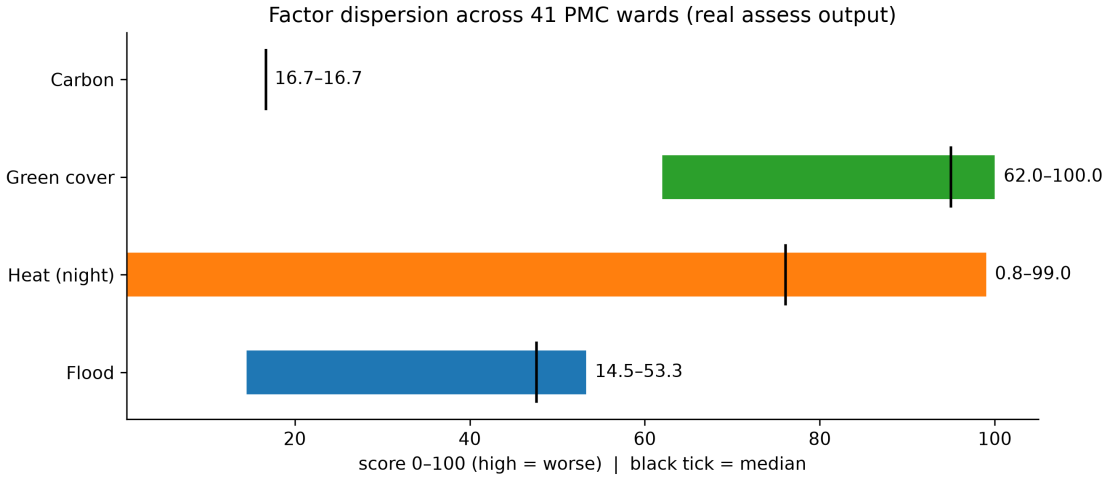


Figure 6. All four factors across 41 wards. Carbon is flat because materials do not vary by location.

8. What the system refuses

Missing data drops a factor out (weights redistribute) — never zero-filled. The narrator cannot alter numbers (prompt rules + tests; one live verdict-softening caught and locked). Unsourced numbers stay labelled (carbon bands, soil group). No neural scorer: one candidate heat model was trained, failed its own bars, and ships nothing.

9. Verification ledger

Claim	Evidence
Flood tracks built cover	r = +0.92, n = 41 real runs
Flood tracks vegetation	r = –0.95
Green tracks vegetation	r = –0.81

Claim	Evidence
Night heat has contrast	city +1.62; r = +0.55 / −0.54
YCEO anchor	+1.139 sampled live
Storm 214 mm	Gumbel KS p = 0.95, CI 191–234
Cement / steel / brick	IFC India Table 16
IGBC / high-rise anchors	700 / 454, primary sources
Census total	4,009,623 unique
Suites	83 passed, 1 skipped (root pytest); e2e Ward 12 all checks; narrative 22/22; engine 23/23; provenance OK

10. Sources

Datasets

Dataset	Source	Link
PMC Tree Census 2019 (17 parts)	OpenCity CKAN package f00d83b8	data.opencity.in
IMD 0.25-degree daily rainfall 1901-2024 (Pai et al. 2014)	IMD Pune, Climate Monitoring group	imdpune.gov.in/cmpg/Griddata
Landsat 8/9 C2L2 surface temperature	USGS via Google Earth Engine	LANDSAT/LC08/C02/T1_L2 (GEE catalog)
MODIS day/night LST, Terra + Aqua (10:30 / 1:30)	NASA via Google Earth Engine	MODIS/061/MOD11A1 + MYD11A1 (GEE catalog)
Dynamic World land cover (9 classes)	Google via Google Earth Engine	GOOGLE/DYNAMICWORLD/V1 (GEE catalog)
ECOSTRESS night LST (70 m)	NASA/JPL via Google Earth Engine	NASA/ECOSTRESS/L2T_LSTE/V2 (GEE catalog)
Yale YCEO urban heat (yearly)	Yale via Google Earth Engine	YALE/YCEO/UHI/UHI_yearly_averaged/v4 (GEE catalog)
GEDI canopy height (RH95)	NASA via Google Earth Engine	LARSE/GEDI/GEDI02_A_002_MONTHLY (GEE catalog)
IFC India materials database (Table 16 GWP)	IFC / EU Eco-cities 2017	edgebuildings.com (IFC India DB methodology report)
PMC ward boundaries (41 wards)	pmc_wards_2025.kml, repo file	repo: pmc_wards_2025.kml

Papers and standards

Reference	Used for	Find it
USDA SCS TR-55 (1986)	Curve numbers, runoff equation	USDA NRCS (TR-55)
Vivekanandan (2022), JWRE journalspub	Gumbel/EV1 choice for Pune	civil.journalspub.info (JWRE Vol.8 No.1)
Das et al. (2022), MAUSAM journal	Gumbel IDF precedent, India	MAUSAM journal
Chow / Maidment / Mays (1988), Applied Hydrology	Annual-maximum method	McGraw-Hill textbook
Stewart & Oke (2012), Bull. Am. Meteorol. Soc.	Local Climate Zones language	AMS journal
Karadumpa et al. (2024), Indian cement plants	Cement factors 0.89–0.94	PubMed 38123770
CEEW steel + aluminium reports	Indian industry factors	ceew.in publications
Akshatha et al. (2025), high-rise study	Mean 454 kgCO2e/m2	ResearchGate 396344543
IGBC Net Zero Carbon rating page	700 kgCO2e/m2 anchor	igbc.in net-zero-carbon-rating
WRI India Pune tree-cover technical note (DOI 10.46830/writn.21.00111)	620 ha loss; census caveats	wri-india.org
Maharashtra Trees Act 1975 + 2021 amendment (PRS Bill No. 8 of 2021)	Felling permission; age-equivalent compensation	prsindia.org; pmc.gov.in tree-protection page

11. Engine reuse and chatbot

Any-region engine. The scorer holds no city knowledge: engine/region.py builds the grid and a validated config scaffold, engine/census.py maps any CSV headers onto the canonical tree index, engine/score.py scores through the live core (imports only, nothing copied). Proven by 23 self-tests on a synthetic region with foreign headers. A new city supplies four things — config, satellite cube, census index, ward polygons — and the same code scores it.

Chatbot narrator. Reads the computed assessment JSON and writes officer-plain prose: verdict first (quoted verbatim), then each factor with its number, conditions with measured effects, the tree ledger, and unknowns stated plainly. Model: Gemini Flash via REST (standard library only); key lives in server-side .env, never in chat, git, or prompts. Verified live: every score, delta, canopy figure, and verdict exact across test wards; one verdict-softening caught on a live run and locked by test. Blocking issues (unknown materials) are mandatory in prose, never buried.

12. Run and reuse

Assess: python -m ml_pipeline.cli assess --ward 12 --built-up 3000 --plot 8000 --materials '[...]' --role municipal_authority --json-out /tmp/out.json. Narrate: python ml_pipeline/report/narrative.py /tmp/out.json (GEMINI_API_KEY in .env). Any other city: engine/ takes a config, a cube, a census index, and ward polygons — the scorer holds no city knowledge.

Known limits. Flood and heat score a 1 km square, not the plot. Daytime surface heat carries no signal here. Carbon bands are house bands with sourced anchors. A plausible-but-wrong input inside 0–100 is undetectable without a second source. No trained model exists here, deliberately.